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Cite this article: Brownstein CD, Griffin CT. 2026 An early burst of skeletal evolution at the origin of dinosaurs. *Proc. R. Soc. B* **293**: 20260102.
https://doi.org/10.1098/rspb.2026.0102

Received: 15 January 2026
Accepted: 4 March 2026

Subject Category:
Palaeobiology

Subject Areas:
palaeontology, evolution

Keywords:
dinosaur, evolutionary radiation, fossil, morphology, phylogenetics

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Electronic supplementary material is available online at https://doi.org/10.6084/m9.figshare.c.8409659.

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An early burst of skeletal evolution at the origin of dinosaurs

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Over 230 million years of Earth's history, dinosaurs became a major terrestrial animal clade and produced one of the most species-rich living tetrapod lineages: birds. Yet, largely because of uncertainty surrounding the phylogeny of early dinosaurs, the tempo and mode of their emergence and initial radiation remain poorly constrained. Here, we reconstruct the initial diversification of dinosaurs through Bayesian tip-dating analyses. Using nine morphological datasets, we estimate that dinosaurs emerged between 250 and 240 Ma, 10 million years before the earliest unambiguous dinosaur fossils. The emergence of the dinosaurs was followed by the rapid appearance and diversification of all major lineages, coinciding with a burst of morphological evolution that peaked in the early Late Triassic. The patterns that we infer are consistent with the expectations under a scenario of evolutionary radiation, in which ecologically disparate lineages rapidly diversify from a single common ancestor. In turn, our results provide a biological explanation for the instability surrounding early dinosaur phylogeny and suggest that the diversity of dinosaurs has been sculpted by multiple rapid radiations following successive mass extinctions in deep time.

1. Introduction

An emerging consensus is that many of the most diverse lineages of animals and plants alive today have originated in evolutionary radiations [1–4]. The past two decades of research on the evolutionary history of *Archosauria* (birds, crocodylians and their extinct relatives) support that the diversity of this clade may have been driven by multiple successive evolutionary radiations, including the initial diversification of ecologically disparate stem and crown archosaur clades in the Triassic [5–19] and the rapid radiation of crown birds after the Cretaceous–Paleogene mass extinction 66.02 million years ago [20–29]. The earliest unambiguous dinosaur fossils appear approximately 230 million years ago in the early Late Triassic [5,6,30,31], implying that dinosaurs postdate the major Early and Middle Triassic archosaur radiation. Early dinosaurs also appear to have been morphologically conservative compared to other archosaurs [10–12,14,16,17,32–34]. These observations suggest at best an indirect association between the initial archosaurian radiation and the origin of the species-rich and ecologically disparate dinosaurs. Nonetheless, the considerable uncertainty surrounding early dinosaur phylogeny [30,35–40] and the roughly simultaneous appearance of large- and small-bodied predatory [38,41–45], omnivorous and herbivorous [46–49] dinosaurs are reminiscent of patterns observed in unambiguous evolutionary radiations and generally match the expectations [3,4] of an evolutionary radiation scenario occurring at the base of *Dinosauria*.

Here, we conduct Bayesian tip-dated phylogenetic analyses of recent, comprehensive morphological character datasets of early dinosaurs and examine patterns of divergence and morphological evolution. We

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demonstrate that, regardless of how the relationships of the major dinosaur clades are inferred, the early evolution of dinosaurs is characterized by a burst of rapid morphological evolution in the Middle to early Late Triassic that followed the origin of the *Dinosauria* approximately 240 million years ago. Rates of morphological evolution are high across the divergences among its major sub-lineages (†*Ornithischia*, †*Sauropodomorpha* and *Theropoda*) but decline within them. These results provide a partial biological explanation for our inability to resolve the base of dinosaur phylogeny and suggest that nested radiations have played a key role in structuring the diversity of even the deepest parts of the avian lineage.

2. Methods

(a) Taxonomy

In this study, we use the conventions described in the PhyloCode [50], which provides for a rank-free classification system based on phylogenetic systematics. Following this system and an emerging convention in the literature [51], we italicize all clade names. We use the dagger symbol (†) to denote extinct taxa.

(b) Taxon–character matrix selection

In addition to the trees produced by Griffin *et al.* [30] from three taxon–character matrices focusing on early dinosaurs [35,36,38], we conducted Bayesian tip-dating analyses on recent datasets exploring early ornithodiran evolutionary relationships [40,52–56]. The Otero & Pol [55] dataset, which is derived from the dataset of Yates [57], is sauropodomorph-focused and tends to support a 'traditional' topology of dinosaurs with a monophyletic *Saurischia* and silesaurids recovered as non-dinosaurs. Li *et al.* [56] is more theropod-focused and derives from various iterations of the dataset of Nesbitt *et al.* [44]. The Cabreira *et al.* [38] and Norman *et al.* [40] datasets tend to support the hypothesis that silesaurids are early diverging ornithischian dinosaurs; the latter dataset derives from the data and results of Müller & Garcia [39]. Garcia & Müller [58] and Müller [54] are subsequent variations of Müller & Garcia [39]. Like Norman *et al.* [40], the Fonseca *et al.* [53] dataset is ornithischian-heavy but largely does not overlap with the matrix of Norman *et al.* [40]. The Baron *et al.* [35] dataset tends to support the *Ornithoscelida* hypothesis of dinosaurian relationships, whereas rescoring of these characters by Langer *et al.* [36] supports the traditional topology of a monophyletic *Saurischia*, as does that of Cabreira *et al.* [38]. However, the Cabreira *et al.* [38] analysis also recovered silesaurids as ornithischians. The nine matrices that we analyse therefore represent six recent, independent datasets that sample both the most recent common ancestor (MRCA) of dinosaurs and all major dinosaur lineages (e.g. sauropodomorphs, theropods) across the Triassic and deep into the Jurassic. This set of morphological matrices includes a wide sample of taxa and characters, including three instances of an iteratively modified matrix (i.e. Norman *et al.* [40], Garcia & Müller [58], Müller [54]), ensuring that the results of our analyses are not driven by a particular scoring scheme, character conception, or phylogenetic hypothesis.

(c) Bayesian tip-dating analyses

We conducted phylogenetic analyses following the methods described by Griffin *et al.* [30] for modelling morphological evolution in a Bayesian phylogenetic context. We conducted all phylogenetic analyses in BEAST v. 2.7.7 [59] using the BEAST implementation of the Fossilized Birth–Death (FBD) model [60] with the Sampled Ancestors and Morphological Models packages. The analysed morphological data matrices use many ordered characters, which BEAST2 is not by default equipped to process as distinct partitions. To address this issue, we partitioned characters both by number of character states and, in the case of ordered characters, through manually coding character states in rate matrices that force stepwise transitions, as in Griffin *et al.* [30]. This produces, for example, a three-state unordered partition as well as a separate three-state ordered partition for the ordered three-state characters. For each partition, we used a Lewis Mk_v model [61] with four gamma-distributed rate categories and a random local clock model.

We assumed the prior on rate changes to have a Poisson distribution with a mean of ln(2) [62], and we placed gamma priors ($\alpha = 0.5396$, $\beta = 0.3819$) on the distribution of each clock rate as demonstrated in prior analyses (e.g. [62]). For tip ages, we used the median of the range of ages known for each taxon, and the ages for each dataset were rescaled to that of the youngest taxon in that dataset, which was set to zero. All absolute taxon age ranges are available in electronic supplementary material, table S1. In almost all cases, the age range stems from dating uncertainty and not a true range, as most taxa are known from a single locality or horizon, and even for those that are not (e.g. *Plateosaurus* spp.), the listed age ranges result from dating uncertainty. Accounting for dating imprecision for each taxon is computationally expensive; to test that the standard usage of median ages (e.g. [30,63]) did not impact our results, for one dataset (Müller [54]), we ran an identical analysis but with a Sampled Ancestors MRCA prior for each taxon, with the dating uncertainty for each taxon given as the range of possible dates in the prior. Because this age range represented dating uncertainty, and not a true range of taxon ages, we set these priors to sample from a Gaussian (rather than uniform) distribution for the age of each taxon. We assigned a broad, weakly informative uniform prior on the origin of the FBD process for each analysis, rather than potentially biasing the analyses with an artificially narrow prior assigned *a priori*. The origin of the FBD process cannot be younger than the oldest taxon in the analysis, so we set the lower bound on the origin prior to slightly (approx. 5 Ma) higher than the age of oldest taxon in the analysis to ensure that the origin is older than the dating error of that taxon. We set the upper bound on this prior to infinity so as to not artificially constrain the age of the root. FBD processes with only weakly informative priors on root age have been shown to estimate node ages well [64], and setting broad priors on tree root age is standard practice (e.g. [30,63]).

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We forced outgroup relationships by creating two topology-only MRCA priors, one for *Archosauria* and one for *Ornithodira*, with the specific taxa included in these MRCA priors dependent on the particular dataset. For example, if †*Euparkeria capensis* was included in a matrix, this taxon was excluded from both the *Archosauria* and *Ornithodira* MRCA priors; if †*Teleocrater rhadinus* was included, this species was only excluded from the *Ornithodira* MRCA prior. To ensure convergence was reached in each analysis, we ran different datasets over different numbers of generations. We ran the Markov-chain Monte Carlo run of Li *et al.* [56] for 40 million generations; Otero & Pol [55], Norman *et al.* [40] and Fonseca *et al.* [53] for 80 million generations; and Garcia & Müller [58] and Müller [54] for 160 million generations. We ran the Müller [54] analysis with age uncertainty for 300 million generations. For every analysis, we used a 50% pre-burn-in and logged a tree every 1000 generations. For each dataset, we ran ten separate chains, evaluating convergence and sampling of each chain with Tracer v. 1.7.2 [65]. We then used LogCombiner v. 2.7.7 to integrate the ten separate chains, with a 10% burn-in of each chain and resampled each chain to log a tree every 4000 generations. This resulted in a total of 90 010 trees [56], 180 010 trees (Otero & Pol [55], Norman *et al.* [40], Fonseca *et al.* [53]) or 360 010 trees (Garcia & Müller [58] and Müller [54]), depending on the length of the analysis. We generated 675 010 trees in the age uncertainty analysis using the Müller [54] dataset. Finally, we created maximum clade credibility trees for each dataset using TreeAnnotator. All phylogenetic analyses were conducted on the Della high-performance computing cluster at Princeton University, and all input and output files are available in the Dryad Digital Repository for this study (https://doi.org/10.5061/dryad.w9ghx3g33).

(d) Filtration and comparison of node and branch rate, age and support values

After collating data on rates of morphological evolution, ages, subtending branch lengths and posterior supports for all nodes in our output time-calibrated phylogenies, we filtered the data to only consist of nodes representing the following lineages: †*Herrerasauria*, †*Ornithischia*, †*Sauropodomorpha*, †*Silesauridae*, *Theropoda*, the closely related †*Lagerpetidae*, and a grade corresponding to nodes representing the divergences between the following major dinosaur clades: †*Herrerasauria*, †*Ornithischia*, †*Sauropodomorpha*, †*Silesauridae* and †*Theropoda*. †*Silesauridae* was identified as the clade including †*Silesaurus* and all taxa more closely related to it than to †*Heterodontosaurus tucki*. In cases where traditional silesaurids such as †*Lewisuchus* are found to be dinosaurs but not placed within †*Silesauridae* as defined, we included the node representing the MRCA of, for example, †*Lewisuchus* and other dinosaurs in the miscellaneous *Dinosauria* bin. We conducted this filtration step to remove nodes associated with outgroups and poorly sampled (fewer than 15 nodes) ingroups (i.e. †*Pterosauria*).

Next, we removed all nodes with ages younger than 170 Ma to account for erroneous long branches produced by tips included to polarize character states, but that fall far outside the targeted region of dinosaur phylogeny, e.g. †*Velociraptor mongoliensis*. We log-transformed all rate data for graphing ease. Using custom scripts in the R package *ggplot2*, we plotted morphological rates, posterior supports and branch lengths through time by clade and by study for the filtered morphological data. We also used *ggplot2* to plot the proportions of taxa sampled by clade, the proportion of missing data in each matrix and the median root-to-tip distances for each maximum clade credibility tree after extracting the latter statistic using TreeStat v. 1.2 (https://tree.bio.ed.ac.uk/software/treestat/). To test for a peak in evolutionary rates through time, we conducted a segmented regression and a Davies' test in the R package *segmented* [66]. Under an early burst pattern of morphological evolution, evolutionary rates would be expected to correlate with time; we tested for this correlation with a Bayesian analysis of correlations in the R package *BayesFactor* [67]; to determine 95% credibility intervals, we sampled from the posterior distribution of the model 10 000 times.

We used generalized additive models (GAMs) to quantify the relationship between time and the natural log of evolutionary rate with the R package *mgcv* [68] with a restricted maximum likelihood smoothing parameter estimation method and $k = 6$, ensuring that there was no evidence this k value was too small ($p > 0.05$ for all models). To assess the certainty and the influence of sampling on the relationship between morphological rates and time, we ran several bootstrapping analyses. After fitting a GAM to the data, we used a residual bootstrap approach, resampling and replacing the original residuals 10 000 times and adding them back to the fitted values, with GAMs fitted to each bootstrap replicate. To investigate how our sample of phylogenetic matrices influenced our results, we employed a matrix-level bootstrap. For each of the 10 000 bootstrap replicates, matrices were resampled with replacement, with all observations combined into a bootstrap dataset. Similar to above, a GAM was then fitted to each bootstrap dataset, with predictions made on a regularly spaced age grid. Similarly, we assessed the influence of individual matrices on our signal with a leave-one-study-out (LOSO) cross-validation analysis, which helps to determine if any individual matrix is driving observed trends. For each matrix, the evolutionary rates derived from the analysis of that matrix were discarded, and a GAM was performed on the remaining data. We plotted the resulting curves alongside the original curve, allowing the influence of each matrix on our reported trend to be observed. To explore the differences in structure between the nine matrices, we also conducted a GAM with each matrix as a factor, fitting a separate curve for each matrix. All output data and R code are available in the Dryad Digital Repository for this study.

3. Results

Our collated dataset of branch lengths, nodal divergence times, posterior support values and rates of morphological evolution (in changes per millions of years) generated from our Bayesian tip-dated phylogenies of nine morphological matrices substantiates several key patterns in the early evolution of dinosaurs (figures 1–3). Because we estimate rates of morphological evolution in dinosaurs and close relatives using multiple morphological matrices with different character and taxon sampling, our results account for the possibility that patterns of morphological evolution are reflective of different species and

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morphological character inclusion [69]. The matrices are the most taxon- and character-rich available for early dinosaurs and relatives and include datasets that broadly sample this phylogenetic interval [35,36] or intensely sample particular lineages, including †*Silesauridae* and 'traditional' (non-silesaurid) †*Ornithischia* [39,40,53,54,58], †*Sauropodomorpha* [38,55] and *Theropoda* [56]. Estimating rates of morphological evolution directly from taxon–character matrices also has the advantage of providing information on rates of change across the skeleton, as opposed to taking select measurements [17,70].

First, we estimate that dinosaurs appeared in the Early to Middle Triassic, between 251.2 and 230 million years ago (figure 1), followed by the rapid appearance of all major dinosaur lineages (†*Ornithischia*, †*Sauropodomorpha*, *Theropoda* and †*Herrerasauria*, a clade of ambiguous dinosaurian affinities) along with the possible dinosaurian clade †*Silesauridae* in the latest Middle to earliest Late Triassic (figure 1). Notably, the recovery of †*Silesauridae* as ornithischian dinosaurs does not substantially contribute to reducing the length of the long branch leading to traditional †*Ornithischia*. This is because, in the majority of our tip-dated phylogenies, most species of silesaurids form a clade exclusive of traditional †*Ornithischia* even when these clades are united within *Dinosauria* (electronic supplementary material, figures S4–S12). Thus, contrary to previous studies [39,40], in these topologies, an ornithischian identity for †*Silesauridae* largely does not help resolve the notorious [35,36,40,71] gap in the early ornithischian record.

Most immediately noticeable in our rate estimates is a pronounced increase in rates of morphological evolution in dinosaurs that peaks between 240 and 220 million years ago before declining into the Jurassic (figure 2A,B). Our segmented regression of evolutionary rate over time suggests that this peak is 233.6 Ma (±2.9; $p < 0.001$), with rate increasing before this time and decreasing after. This is supported by a Davies' test for change in slope, which suggests a change from positive to negative slope at 233.48 Ma ($p = 0.0003$)—this peak is roughly coincident with the so-called dinosaur diversification event and the Carnian Pluvial Episode (e.g. [72–74]). Our Bayesian analysis of correlations showed strong evidence for a correlation between evolutionary rate and time (rho = 0.333; 95% credibility interval = 0.250–0.400; Bayes factor ≈ 6.5×10^{12}), which is expected under a pattern of time-dependent shifting rates (i.e. early burst) but not under constant evolutionary rates. Bootstrapping and LOSO analyses demonstrate that this pattern is robust and not driven solely by any particular combination of morphological matrices in our study (electronic supplementary material, figure S1). Our GAM strongly suggested a nonlinear relationship between age and evolutionary rate (effective degrees of freedom (EDF) = 4.34, $p < 0.001$), with 15.7% of the deviance in evolutionary rate explained by age. Factoring by matrix increased the deviance explanation considerably (66.9%), suggesting that rates across time differ between matrices. We explicitly tested this with a likelihood-ratio test, which strongly favoured the matrix-specific model (ΔEDF ≈ 19.8, Δdeviance = 6.34, χ^2 test, $p < 2 \times 10^{-16}$). In this matrix-specific model, the smooth terms for six matrices were significant (EDF = 1.6–4.1; $p < 0.05$), suggesting an age-dependent structure to evolutionary rates in these trees, and the plotted values suggest high rates early in evolutionary history consistent with an early burst pattern (electronic supplementary material, figures S1 and S2). However, the smoothing terms of the Fonseca *et al.* [53], Norman *et al.* [40] and Li *et al.* [56] trees were not significant (EDF = 1.00; $p > 0.05$), indicating no variance in evolutionary rate across time and an essentially 'flat' curve (electronic supplementary material, figure S2). We interpret this divergence from the other six trees as largely driven by the taxon sampling of Li *et al.* [56] and Fonseca *et al.* [53], which heavily sample Late Triassic to Jurassic theropods and ornithischians, respectively (electronic supplementary material, figure S3A,B), instead of sampling the early divergences among dinosaur clades where this early burst of morphological evolution is expected to have occurred. The wide confidence intervals for the rate curves generated from the tip-dated phylogenies inferred using the Li *et al.* [56], Fonseca *et al.* [53] and Norman *et al.* [40] datasets do not preclude more complex structure in evolutionary rate over time not captured by these models (electronic supplementary material, figure S2). The iteratively modified matrix of Norman *et al.* [40], first by Müller [54] and then by Garcia & Müller [58], demonstrates this pattern: whereas Norman *et al.* [40] returns a 'flat' curve, both Garcia & Müller [58] and Müller [54] infer more complex relationships between evolutionary rate and time, including high early evolutionary rates (electronic supplementary material, figure S2).

Notably, if the fossil record is interpreted uncritically with respect to the age of dinosaurian origins—that is, if the oldest unambiguous dinosaurian fossils are taken to record the absolute age of the dinosaurian common ancestor (approx. 233 Ma)—this initial increase in rates would necessarily be much higher, as the same number of changes in character states would need to occur over a shorter period of time to produce the same inferred history of morphological evolution. Our inferred initial spike in dinosaurian evolutionary rate is therefore a conservative estimate. Rates of rapid morphological evolution are concentrated among nodes representing divergences among the major dinosaur clades (†*Herrerasauria*, †*Ornithischia*, †*Sauropodomorpha*, †*Silesauridae*, *Theropoda*) and within †*Herrerasauria* (figures 2B and 3). Rates of morphological evolution in the three dinosaurian lineages that persist through the Triassic–Jurassic mass extinction also decline from an early peak in the Carnian, although *Ornithischia* displays an elevated rate of morphological evolution into the Early Jurassic (figure 2). Although there is a high degree of variance, rates of variation across the earliest divergences in *Dinosauria* and in the exclusively Triassic †*Herrerasauria* are higher than any of the dinosaurian lineages that persisted through the Triassic–Jurassic mass extinction (figure 3A). This suggests that an increase in morphological evolutionary rate was largely restricted to the inter-lineage relationships among the major dinosaur clades. Furthermore, morphological rates of evolution show more temporal than lineage-specific structure; although rates of morphological evolution are highest at nodes representing the successive divergences among major dinosaur lineages (figure 3A), rates of morphological evolution vary more across morphological matrices (electronic supplementary material, figure S2) than clades (figure 3). Finally, the inferred high rates of morphological evolution that characterize the earliest divergences in dinosaurs are not solely attributable to taxon sampling. Although a peak in node density does occur between 225 and 250 million years ago, two additional peaks occur between 225 and 200 million years ago, whereas rates of morphological evolution estimated for the examined nodes fall dramatically across this interval (figure 2).

There are no clear signals of reductions in successive branch length values (in millions of years) across dinosaurs that correspond to the age of this burst in morphological evolution. On average, successive divergences among dinosaur clades

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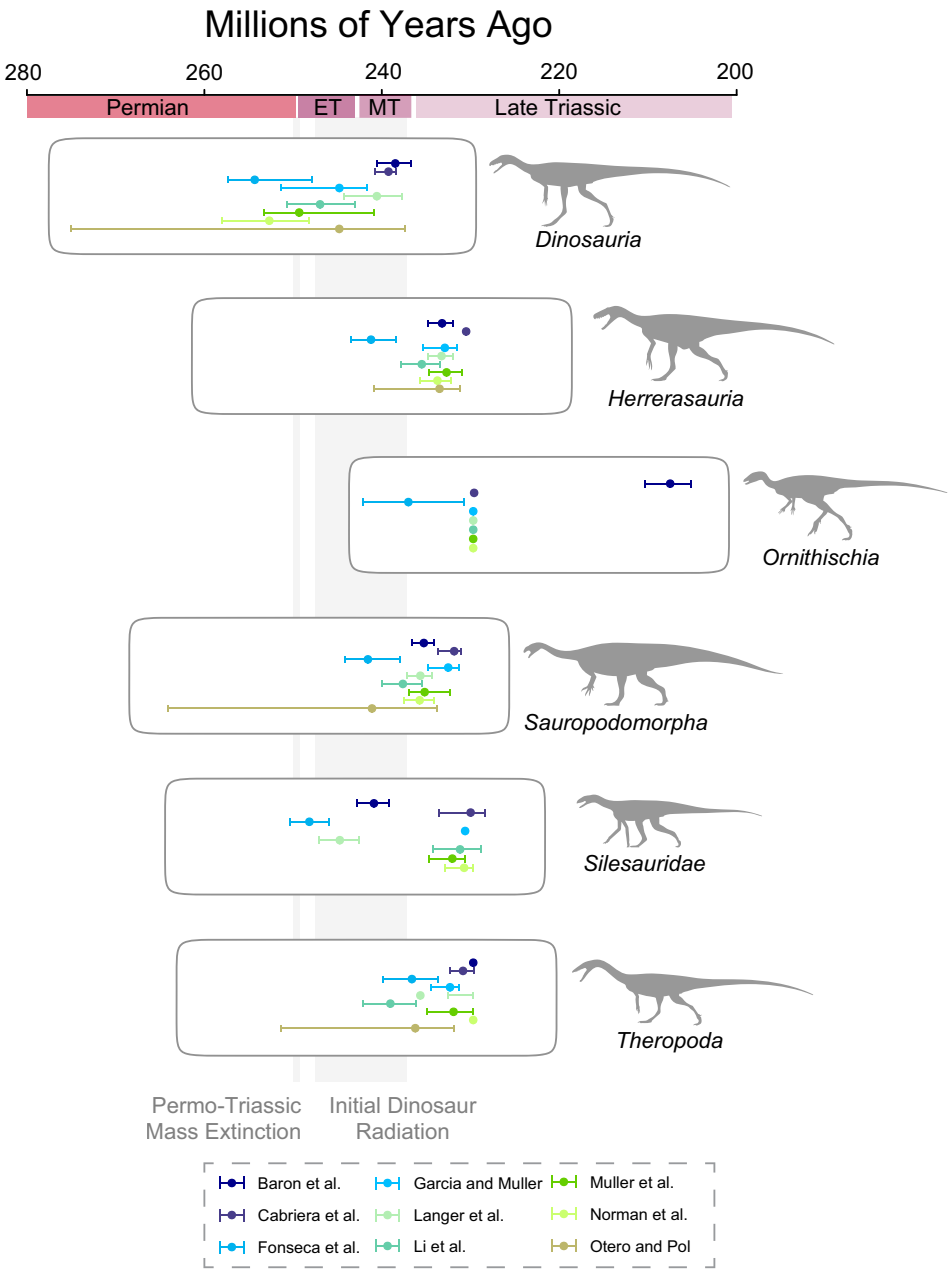


Figure 1. Estimating the age of dinosaurs. The plot shows median ages (dots) and 95% highest posterior density intervals (bars) estimated for six lineages across our Bayesian tip-dated phylogenies from nine different morphological matrices.

in our tip-dated phylogenies occur every five million years (figure 3B). However, ornithischian dinosaurs deviate from other dinosaurs in the tempo of lineage divergence. Ornithischians show the highest variance in branch lengths among dinosaurs and closely related archosaurian lineages (e.g. †*Lagerpetidae*) and also display a clear drop in average branch length that corresponds to the Triassic–Jurassic boundary (figures 2C, 3B and 4A). The right skew (figure 3B) in ornithischian branch lengths is due to the long branches leading to two clades: †*Pisanosaurus* + traditional †*Ornithischia*, and traditional †*Ornithischia* (electronic supplementary material, figures S4–S9). Indeed, when †*Pisanosaurus* is found to be a member of †*Silesauridae* (as in iterations of the matrix of Baron *et al.* [35,71]), the age of †*Ornithischia* is found to be Late Triassic, suggesting the common ancestor of all traditional †*Ornithischia* originated long after those of *Theropoda*, †*Herrerasauria* and †*Sauropodomorpha*.

Trends in posterior support for nodes are also variable across different dinosaur lineages. As expected based on previous analyses that find low support for any particular hypothesis of early dinosaur interrelationships [36,37], posterior support values for nodes representing divergences among major dinosaur clades, nodes in †*Herrerasauria*, and nodes in †*Sauropodomorpha* dip between 240 and 230 million years ago (figure 4B). In contrast, nodal support for clades in †*Silesauridae* and †*Lagerpetidae* peaks during this interval (figure 4B). A peak in nodal support in †*Ornithischia* occurs in the latest Triassic between 200 and 220 million years ago (figure 2D), which is probably an artefact of the low number of divergences that occur in this clade during this interval. Subsequent to the Triassic–Jurassic boundary, average nodal support in †*Ornithischia* drops by 20–30% relative to this Late Triassic peak (figure 4B).

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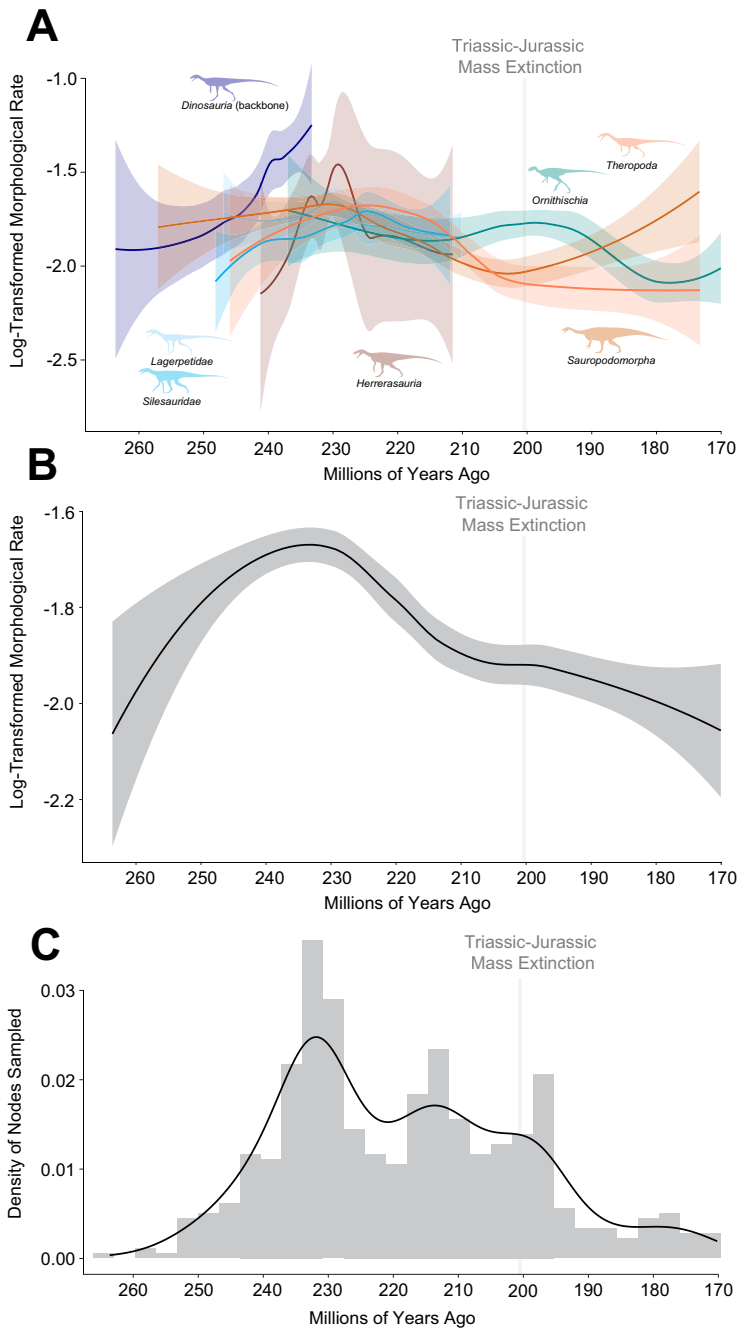


Figure 2. A burst of morphological evolution at the dawn of the dinosaurs. Graphs show best fit curves and 95% confidence intervals for rates of morphological evolution by clade (A) and pooled (B) through time. In (B), the burst of morphological evolution associated with the origin and early diversification of dinosaurs is clearly observable. (C) A density plot of ages for nodes sampled. Grey bars indicate the Triassic–Jurassic mass extinction.

4. Discussion

Our analyses provide consilience surrounding the emergence and tempo of morphological evolution in the earliest dinosaurs. By accounting for uncertainty surrounding the relationships of the major dinosaur clades by analysing multiple morphological matrices that produce conflicting phylogenies of *Dinosauria*, we demonstrate that dinosaurs likely appeared in the late Early to Middle Triassic (figure 1), which predates the earliest unambiguous dinosaur fossils [30,31,41,44,75–79] but is comparable to the ages of the oldest ichnofossils assignable to dinosaurs. or very closely related taxa [80,81], as well as hints in the body fossil record [82] and other recent quantitative estimates [30,83,84]. The age of *Dinosauria* is historically contentious owing to the (i) high lineage diversity and (ii) wide distribution of dinosaurs upon their initial appearance in the fossil record between 230 and 233 million years ago [30,31,41,42,78,79]. A recent analysis of dinosaur historical biogeography demonstrated that, given an earlier age for *Dinosauria*, patterns of dinosaur dispersal support the hypothesis that arid bands crossing the low latitudes of Pangaea acted as a barrier to dispersal across the supercontinent [30]. Historical biogeographic analyses of other archosauromorph clades that incorporate phylogenetic information also find evidence that the arid band acted as a dispersal barrier to terrestrial [85,86] and semiaquatic [87] lineages and was relaxed by the Carnian Pluvial Episode, a period of increased rainfall that occurred between approximately 234 and 232 million years ago [73,88,89]. The Early to Middle Triassic age for *Dinosauria* that we estimate is consistent with this scenario of alternate dry and wet intervals restricting and permitting the dispersal of

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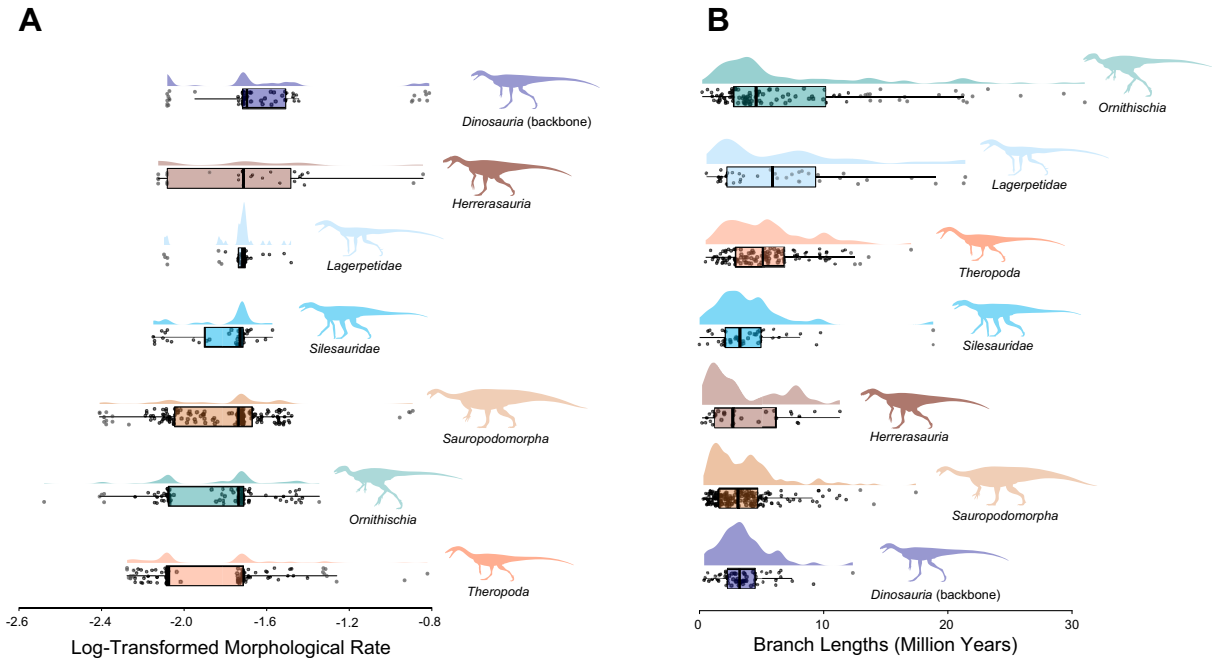


Figure 3. Morphological rates and branch lengths for selected clades. Box and whisker plots for (A) rates of morphological evolution and (B) branch lengths in millions of years for selected clades.

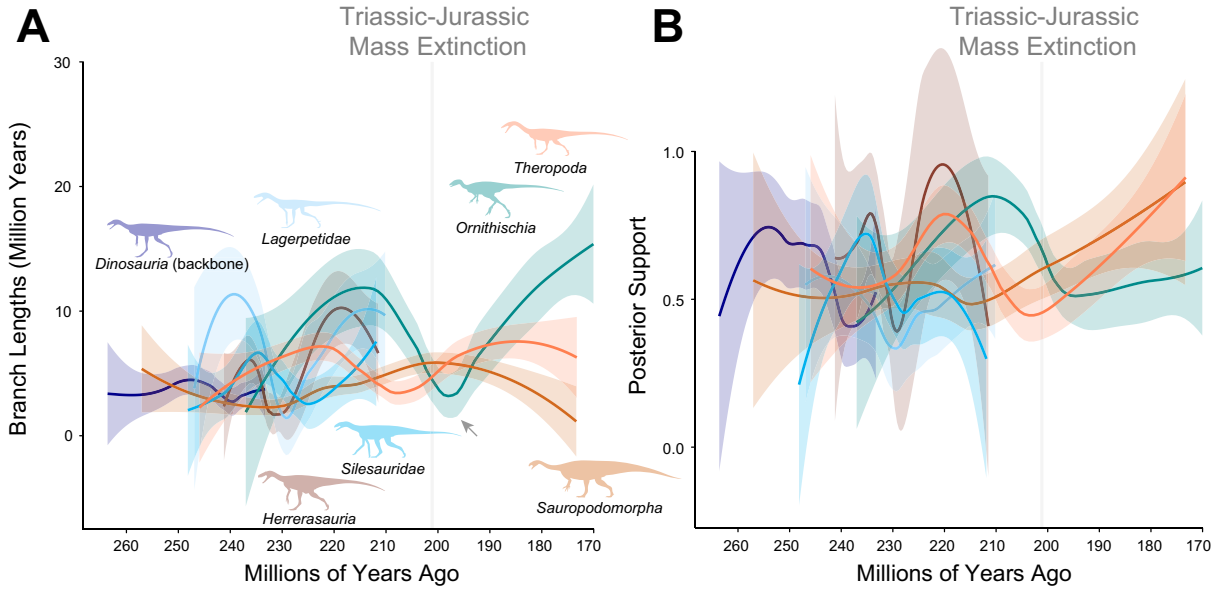


Figure 4. Branch lengths and posterior support through time. Graphs show best-fit curves and 95% confidence intervals for branch lengths by clade (A) and posterior support by clade (B) through time. Grey bars indicate the Triassic–Jurassic mass extinction. Arrow points to the post-Triassic drop in ornithischian rates.

dinosaurs and other reptiles across Pangea. This scenario entails the testable prediction that unambiguous dinosaurian fossils will be recovered from Early or Middle Triassic sediments, most likely from southern Pangea, and will continue to be absent from coeval northern sediments. Indeed, if either silesaurids or †*Nyasasaurus* are dinosaurs, then this may already be the case [75,82,90,91].

Across the tip-dated phylogenies that we inferred, a burst of rapid morphological evolution occurs among the earliest divergences among the major dinosaur lineages—†*Herrerasauria*, †*Ornithischia*, †*Sauropodomorpha*, *Theropoda* and potentially †*Silesauridae* (figures 2 and 3). This early burst of morphological evolution coincides with the origins of ecologically disparate lineages of dinosaurs (e.g. small and large carnivores [38,41–44], disparate herbivores [45–49]) and matches patterns of morphological evolutionary rate increases in other putative ancient adaptive radiations, including archelosaurs in the aftermath of the Permo–Triassic mass extinction [18,92] and stem mammals during the Middle Jurassic [69]. Critically, our analyses substantiate three key patterns of evolution in early dinosaurs that match expectations of adaptive radiation [1–3]. First, our analyses suggest that, regardless of their precise phylogenetic relationships, the first sampled nodes of all major dinosaurian clades originated within five million years of the origin of *Dinosauria*. Second, the appearance of these ecologically disparate [41,46,47,93,94] subclades coincided with an early burst of morphological evolution that preceded the origin of distinct body plans. Third, rates of morphological evolution in dinosaurs declined precipitously in the latest Triassic and potentially into the earliest Jurassic, even though overall dinosaurian taxonomic diversity increased during this interval [5,6,17,95]. The consilience

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between patterns of evolution in early dinosaurs and expected trends under a scenario of evolutionary radiation suggests that the origin and early diversification of dinosaurs in the Middle to earliest Late Triassic (Anisian to Carnian Stages) is an example of this macroevolutionary phenomenon.

What triggered this apparent radiation of early dinosaurs? Our results suggest that several hypothesized drivers do not closely match the age of the initial dinosaur diversification. The origin of endothermy in dinosaurs is poorly constrained, and different studies either reconstruct its origin deep among Triassic archosaurs or in Early Jurassic dinosaurs [96–98]. An expansion of dinosaurian geographic distribution also appears to have, at best, been an indirect or secondary driver of dinosaur diversification, as abrupt Triassic climate change events such as the Carnian Pluvial Event [30,73,89,99] and cooling driven by volcanism in the Central Atlantic Magmatic Province [100,101] postdate both the origin of *Dinosauria* and the rapid increase in morphological evolution along branches representing the successive divergences of the major dinosaur lineages (†*Ornithischia*, *Theropoda*, †*Sauropodomorpha*, etc.) that we detect. More compelling is simply the presence of ecological opportunity [3,4] in the aftermath of the End Permian extinction, which appears to have driven the rapid diversification of stem- and crown-group reptiles in terrestrial and aquatic ecosystems [7,12,15,18,19,92,102–106].

An ecological opportunity hypothesis for the initial dinosaur radiation need not invoke the oft-associated suggestion that dinosaurs outcompeted other reptiles for ecological dominance [6,17,107], as often classically proposed [108,109], and also matches evidence from the fossil record that large tetrapod faunal turnover directly preceded the initial diversification of dinosaurs in southern Pangaea [110]. Although our results do not directly inform the cause of the initial diversification of dinosaurs, they are congruent with a sudden burst of morphological innovation that matches expected patterns from studies of living clades. For example, the exceptionally diverse cichlid species flocks of the African rift lakes co-occur with cyprinid and siluriform fishes that also show adaptive radiation-like evolutionary dynamics [111,112]. The fossil record of Lake Victoria, which houses more than 500 species of cichlids that originated in fewer than 20 000 years [113–115], records that cichlids were neither the first lineage of fishes to enter the lake nor were they immediately ecologically dominant upon their initial appearance [116]. Instead, the diversification of cichlids in African rift lakes and elsewhere in the southern hemisphere might best be understood in terms of the nested origination of key innovations and exploitation of ecological opportunity rather than intense competitive exclusion dynamics with other fish clades [117,118].

We also highlight that the diversification of dinosaurs appears to be consistent with a nested radiation (*sensu* [119]). Indeed, the fossil record of Triassic reptiles demonstrates that stem archosaurs [9,16,18,92,120], and possibly stem archosaur [92,102] marine reptile clades including ichthyosaurs [105,106] show evidence for exceptionally rapid radiation after the Permo–Triassic mass extinction. If this is the case, the initial radiation of dinosaurs is nested within a larger radiation of stem and crown archosaurs that initiated around the Permian–Triassic boundary.

Finally, our results imply that the evolutionary history of dinosaurs has been driven by punctuated bursts of morphological evolution that occurred sporadically through the 240-million-year evolutionary history of the clade [6,20,23,25,28,95,121]. At least two of the three major mass extinctions that have occurred over this interval appear to be the most direct drivers of bursts of diversification in dinosaurs. In contrast, we do not detect a shift in rates of morphological evolution in dinosaurs following the Triassic–Jurassic mass extinction, even though the tip-dated phylogenies that we analyse sample many nodes and tips that date well into the Jurassic and thus ought to capture morphological evolutionary rates well into this geological period (figure 2; electronic supplementary material, figure S3). The most obvious explanation for the apparent absence of a response to the Triassic–Jurassic mass extinction detected in our analyses is incomplete sampling of relevant dinosaur lineages (although see [122]). For example, time-calibrated phylogenies of *Coelurosauria*, the clade of theropods that includes birds and most feathered dinosaurs, suggest this group rapidly diversified in the Middle Jurassic, approximately 170 million years ago [123,124]. The more inclusive theropod clade *Averostra* also appears to have undergone rapid radiation in the Early and Middle Jurassic [125]. Most phylogenies of dinosaurs are time-calibrated *a posteriori* from trees built using parsimony analysis [126] and often are calibrated using minimum-branch-length methods. As such, the application of Bayesian methods might push the ages of these theropod radiations closer to the Triassic–Jurassic mass extinction [127]. A burst of morphological evolution and increased lineage diversification in dinosaur clades following the Triassic–Jurassic mass extinction is perhaps expected, although previous studies (e.g. [122]) that examined the tempo of early dinosaur diversification using phylogenies have also found limited evidence for morphological rate or diversification shifts following the Triassic–Jurassic boundary. These comparisons underscore the power of phylogenetics for revealing key episodes in the history of iconic lineages like dinosaurs that would be obscured by an uncritical reading of the fossil record.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. All data and code to reproduce the analyses and results in this paper are in the electronic supplementary material or the associated online Dryad repository [128].

Electronic supplementary material is available online [129].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. C.D.B.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing—original draft, writing—review and editing; C.T.G.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing—original draft, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. C.D.B. is supported by the Yale Training Program in Genetics (NIH grant: 5T32GM148332-02).

Acknowledgements. We thank B. Wynd for his help and advice regarding Bayesian phylogenetic methods and BEAST 2, and S. Nesbitt for discussion on the ornithischian ghost lineage. The phylogenetic analyses were performed on the Della computing cluster through the Princeton Research Computing resources at Princeton University, led by the Princeton Institute for Computational Science and Engineering (PICSciE)

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and Research Computing. We thank T. Simões for the use of his BEAST 2 module on Della and for discussions on evolutionary radiations. All silhouettes used in figures are by Scott Hartman from phylopic.org (CCBY 3.0).

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